Air University Islamabad
Department of Creative Technologies

AI303-Computer Vision

Semester: Spring 2026 **Date**: April 27th, 2026
Instructor: Ms. Hina Rashid

Assignment 1: Radiometric Pre-processing & Noise Mitigation

Duration: 1 Week | **Weight:** Part of 10% | **Focus:** CLO-1, CLO-2

Objective: To clean raw medical and remote sensing data by adjusting the system's response to noise and light

1. Data Assignment by Groups

- **Groups 1–8:**3D Brain Tumor MRI Slices
- **Groups 9–10:** Canine Heart MRI slices (density-based tissue analysis)
- **Groups 11–12:** Night-Time Lights of the World (Remote sensing/GIS data)
- **Groups 13–15:** Industrial PCB component images or Red Blood Cell images

2. Tasks

- **Batch Processing:** Implement a Python script to load the entire **folder of images**.
- **Radiometric Resolution:** Identify the bit-depth of your images (e.g., 8-bit vs. 16-bit) to avoid **False Contouring** during processing
- **Filtering:** Implement **Gaussian, Mean, and Median (Appropriate) filters** to remove sensor noise. Quantify the restoration using **PSNR and SSIM** metrics
- **Anti-Aliasing Challenge:** If downsampling your data, you must apply a **Gaussian pre-filter** to avoid the **Checkerboard Effect**

3. Deliverables

- Python Notebook (.ipynb), "Before/After" dataset folder, and a technical report documenting the **Point Spread Function (PSF)** adjustment

Dataset Repository Links for Assignment 1:

- **Brain Tumor/Medical:** TCIA Medical Imaging Archive
- **Heart/Canine MRI:** Open-Source Medical Datasets (Kaggle)
- **GIS/Satellite:** Night-Time Lights of the World (NOAA)

Assignment 2: Mid-Level Structural Representation & Masking

Duration: 1 Week | **Weight:** Part of 10% | **Focus:** CLO-3

Objective: To extract boundaries and simplify complex shapes into "ready-to-use" geometric primitives.

1. Tasks

- **Segmentation (Masking):** Apply **Canny or Sobel Edge Detection** to create a binary mask of the tumor, heart chamber, or satellite terrain

- **Morphological Cleaning:** Use **Erosion, Dilation, Opening, and Closing** to remove binary artifacts and "salt and pepper" noise from the masks
- **Boundary Representation:**
- Generate an **8-directional chain code** for the primary object
- Derive the **First Difference** and **Shape Number** to ensure the descriptor is **rotationally invariant** (essential for tilted scans)
- **Computational Geometry:** Implement the **Graham Scan** or **Jarvis March** algorithm to find the **Convex Hull** of the detected region

2. Deliverables

- Code files and a report containing **Chain Code tables** and **Convex Hull overlays** on the original data

Assignment 3: Statistical Texture Analysis & Traditional Classification

Duration: 1 Week | **Weight:** Part of 10% | **Focus:** CLO-1, CLO-4

Objective: To characterize tissue/regions using statistical properties rather than just geometry

1. Tasks

- **GLCM Computation:** For your masked regions (from A2), compute the **Gray-Level Co-occurrence Matrix**
- **Statistical Descriptors:** Extract **Energy** (uniformity), **Entropy** (randomness), and **Contrast** (local variation
- **Geometric Features:** Calculate the **Area**, **Centroid**, **Perimeter**, and **Circularity** of the object
- **Traditional Classification:** Use these features to label regions (e.g., "Malignant vs. Benign" or "Urban vs. Rural")

2. Deliverables

- Feature vector CSV files and a classification accuracy report

Final Semester Project: The Deep Vision Pipeline

Duration: 10 Days | **Weight:** 05% | **Focus:** CLO-5

Objective: To develop a high-level application using **Deep Neural Networks** for object detection and semantic understanding

1. The Challenge

Integrate your findings from Assignments 1–3 into an end-to-end pipeline: **Acquisition->Pre-processing->Segmentation->Description-> Recognition**

2. Technical Requirements

- **Applying Models:** Implement a deep learning architecture (e.g., **CNN** for classification or **YOLO** for localization) on your assigned dataset
- **Advanced Topics:** Groups must choose one "Research Challenge" for bonus marks:
- **3D Transformer Integration:** Apply a transformer block for voxel-to-voxel dependencies
- **Specular Mitigation:** Identify and remove **specular reflections** (shiny areas) that interfere with segmentation in medical videos
- **Evaluation:** Provide a complete confusion matrix and precision/recall analysis.

3. Final Deliverables

- **Code:** Full GitHub repository link
- **Data:** Final annotated dataset used for training/testing.

- **Report:** A professional 15-page research paper including a **Literature Review** and **Methodology and findings**

Good Luck!